MTH 132 Midterm 2013

Name______________________________ Date___________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Use the given conditions to write an equation for the line in slope–intercept form.

1) Slope = –3, passing through (6, 6) 1) ________

A) $y - 6 = -3x - 6$ B) $y - 6 = x - 6$ C) $y = -3x + 24$ D) $y = -3x - 24$

2) Slope = 4, passing through (7, 3) 2) ________

A) $y = 4x - 25$ B) $y - 3 = 4x - 7$ C) $y = 4x + 25$ D) $y - 3 = x - 7$

Find the equation of the least squares line.

3) The paired data below consist of the test scores of 6 randomly selected students and the number of hours they studied for the test. 3) ________

Hours (x)	5	10	4	6	10	9
Score (y)	64	86	69	86	59	87

A) $y = 33.7 + 2.14x$ B) $y = 67.3 + 1.07x$
C) $y = 33.7 - 2.14x$ D) $y = -67.3 + 1.07x$

Write a cost function for the problem. Assume that the relationship is linear.

4) A moving firm charges a flat fee of \$40 plus \$35 per hour. Let C(x) be the cost in dollars of using the moving firm for x hours. 4) ________

A) $C(x) = 40x - 35$ B) $C(x) = 35x - 40$ C) $C(x) = 35x + 40$ D) $C(x) = 40x + 35$

Solve the system.

5) 5) ________
$$x + y = -2$$
$$x - y = 18$$

A) {(8, –10)} B) {(8, 10)} C) {(2, –10)} D) {(2, 8)}

Solve the system of two equations in two unknowns.

6) 6) ________
$$x + 6y = -5$$
$$-4x + 7y = 20$$

A) (–5, 0) B) (–4, –5) C) (5, –1) D) No solution

Find the slope of the line passing through the given pair of points.

7) (3, 4) and (3, −8) 7) ________

A) $-\frac{2}{3}$ B) Undefined C) 0 D) 2

8) (−4, 1) and (−9, 1) 8) ________

A) Not defined B) 0 C) $-\frac{2}{13}$ D) $-\frac{2}{5}$

9) (5, −5) and (−7, −3) 9) ________

A) $-\frac{1}{6}$ B) 4 C) − 6 D) $\frac{1}{6}$

10) (6, 5) and (−4, 5) 10) ________

A) 5 B) Undefined C) − 1 D) 0

The sizes of two matrices A and B are given. Find the sizes of the product AB and the product BA, whenever these products exist.

11) A is 1 × 3, and B is 3 × 1. 11) ________

A) AB is 1 × 1; BA is 3 × 3. B) AB is 3 × 3; BA is 1 × 1.

C) AB does not exist; BA is 3 × 3. D) AB is 1 × 1; BA does not exist.

Solve the problem.

12) 12) ________

Let $A = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 6 \\ 3 & 1 \end{bmatrix}$. Find A − B.

A) $\begin{bmatrix} 0 & 6 \\ -3 & -2 \end{bmatrix}$ B) $\begin{bmatrix} 0 & -6 \\ 3 & 2 \end{bmatrix}$ C) $\begin{bmatrix} -1 \end{bmatrix}$ D) $\begin{bmatrix} -2 & 6 \\ 9 & 4 \end{bmatrix}$

Find the matrix product, if possible.

13) $\begin{bmatrix} 1 & 3 & -2 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix}$ 13) ________

A) $\begin{bmatrix} 3 & -6 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ B) $\begin{bmatrix} -3 & -3 \\ 6 & 9 \end{bmatrix}$ C) $\begin{bmatrix} -3 & -3 \\ 9 & 6 \end{bmatrix}$ D) Does not exist

Solve the problem.

14) Let $A = \begin{bmatrix} -3 & 2 \\ 0 & 2 \end{bmatrix}$. Find 4A. 14) _______

A) $\begin{bmatrix} -12 & 2 \\ 0 & 2 \end{bmatrix}$
B) $\begin{bmatrix} 1 & 6 \\ 4 & 6 \end{bmatrix}$
C) $\begin{bmatrix} -12 & 8 \\ 0 & 8 \end{bmatrix}$
D) $\begin{bmatrix} -12 & 8 \\ 0 & 2 \end{bmatrix}$

15) Let $A = \begin{bmatrix} 2 & 3 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ -1 & 6 \end{bmatrix}$. Find 2A + B. 15) _______

A) $\begin{bmatrix} 4 & 10 \\ 1 & 10 \end{bmatrix}$
B) $\begin{bmatrix} 4 & 14 \\ 2 & 20 \end{bmatrix}$
C) $\begin{bmatrix} 4 & 7 \\ 3 & 10 \end{bmatrix}$
D) $\begin{bmatrix} 4 & 10 \\ 3 & 14 \end{bmatrix}$

16) Let $A = \begin{bmatrix} -5 & 1 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 2 \\ 3 & 2 \end{bmatrix}$. Find A + B. 16) _______

A) $\begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$
B) $\begin{bmatrix} 16 \end{bmatrix}$
C) $\begin{bmatrix} 3 & 4 \\ 3 & 7 \end{bmatrix}$
D) $\begin{bmatrix} 1 & -3 \\ -2 & -3 \end{bmatrix}$

Find the matrix product, if possible.

17) $\begin{bmatrix} 3 & -2 & 1 \\ 0 & 4 & -1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ -2 & 3 \end{bmatrix}$ 17) _______

A) $\begin{bmatrix} 9 & -6 & 3 \\ -6 & 16 & -5 \end{bmatrix}$
B) $\begin{bmatrix} 9 & -6 \\ -6 & 16 \\ 3 & -5 \end{bmatrix}$
C) $\begin{bmatrix} 9 & 0 \\ 0 & 12 \end{bmatrix}$
D) Does not exist

Find the inverse, if it exists, for the matrix.

18) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ -2 & 3 & 1 \end{bmatrix}$ 18) _______

A) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ -5 & 4 & 1 \end{bmatrix}$
B) No Inverse
C) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ -5 & 4 & -1 \end{bmatrix}$
D) $\begin{bmatrix} 1 & -1 & 0 \\ 3 & -2 & 1 \\ -5 & 4 & -1 \end{bmatrix}$

Solve the problem.

19) A digital pen manufacturer has decided to come out with a new and affordable digital pen . The fixed cost for the production of this new digital pen line is \$16,600 and the variable costs are \$68 per digital pen . The company expects to sell the digital pens for \$159. Formulate a function C(x) for the total cost of producing x new digital pens and a function R(x) for the total revenue generated from the sales of x digital pens. 19) _______

A) C(x) = 16600 + 159x; R(x) = 68x
B) C(x) = 16600 + 68x; R(x) = 159x
C) C(x) = 16,668; R(x) = 159
D) C(x) = 68x; R(x) = 159x

20) There were 570 people at a school fundraising concert. The admission price was \$2 for adults and \$1 for children. The admission receipts were \$830. How many adults and how many children attended? 20) _______

A) 207 adults, 363 children
B) 155 adults, 415 children
C) 310 adults, 260 children
D) 260 adults, 310 children

21) Sweet Honey Bakery serves desserts. One serving of ice cream and two servings of blueberry pie provides 820 calories. Three servings of ice cream and two servings of blueberry pie provides 1300 calories. Find the caloric content of each item. 21) _______

A) Serving of frozen yogurt: 240 calories
Serving of apple cobbler: 290 calories
B) Serving of frozen yogurt: 290 calories
Serving of apple cobbler: 240 calories
C) Serving of frozen yogurt: 252 calories
Serving of apple cobbler: 284 calories
D) Serving of frozen yogurt: 234 calories
Serving of apple cobbler: 293 calories

22) After two years on the job, a therapist's salary was \$60,000. After seven years on the job, her salary was \$72,500. Let y represent her salary after x years on the job. Assuming that the change in her salary over time can be approximated by a straight line, give an equation for this line in the form y = mx + b. 22) _______

A) y = 12,500x + 60,000
B) y = 2500x + 60,000
C) y = 12,500x + 35,000
D) y = 2500x + 55,000

23) FedUps delivers packages which cost \$2.00 per package to deliver. The fixed cost to run the delivery truck is \$234 per day. If the company charges \$5.00 per package, how many packages must be delivered daily to make a profit of \$42? 23) _______

A) 33 packages
B) 117 packages
C) 78 packages
D) 92 packages

24) A shoe company will make a new design of walking shoe. The fixed cost for the production will be \$24,000. The variable cost will be \$31 per pair of shoes. The shoes will sell for \$106 for each pair. How many pairs of shoes will have to be sold for the company to break even on this new line of shoes? 24) _______

A) 321 pairs B) 775 pairs C) 75 pairs D) 227 pairs

25) Let the supply and demand functions for gourmet hot chocolate packs be given by 25) _______

$$p = S(q) = \frac{5}{4}q \quad \text{and} \quad p = D(q) = 70 - \frac{3}{4}q,$$

where p is the price in dollars and q is the number of batches.. Find the equilibrium quantity and the equilibrium price.

A) Equilibrium quantity: 35
Equilibrium price: \$43.75

B) Equilibrium quantity: 35.00
Equilibrium price: \$28

C) Equilibrium quantity: 28
Equilibrium price: \$35.00

D) Equilibrium quantity: 43.75
Equilibrium price: \$35

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

26) In one or more paragraphs, explain what it means to "break even" in business. 26) _______________